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# Causal Inference With Randomization

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Science asks, “If I do this, what will happen?”

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Richard Feynman

Science allows us to make better choices: Doctors use scientific results to choose how to treat patients, engineers choose materials based on the scientific knowledge of the properties of those materials. At the heart of these decisions is the conceptualization of causality: In order to determine what to do, we wish to evaluate what we believe will happen if we do the action. Normally, we also want to compare what would happen if we take one potential action, relative to what may happen if we take a different action. *To treat or not to treat? That is the question.* Unlike traditional forms of inference, which only work to establish associations among what one has already observed, Causal Inference allows one to investigate and compare these future potential outcomes; i.e. what would happen if you actively intervened to change something.

This chapter provides an overview of Causal Inference concepts and basic techniques which are deployed in cases where randomization is possible. Along with its successor on Causal Inference Without Randomization, this material serves as our initial reference point, which we will build bridges to and from in the subsequent chapters on Reinforcement Learning. To establish a solid foundation for each bridge, we include some signpost indicators of future crossover material between Causal Inference and Reinforcement Learning as well; readers unfamiliar with Reinforcement Learning should not be alarmed if these are unknown, as they will be developed more explicitly in later chapters.

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### 2.1 Causal Treatment Effects

Suppose an asthma patient comes to a doctor for treatment. The doctor knows of two potential medications that might help. The doctor therefore has two potential actions she could take:

1. Action  $a^{(1)}$ : Prescribe drug 1
2. Action  $a^{(2)}$ : Prescribe drug 2

Suppose she chooses  $a^{(1)}$ , prescribing drug 1, and observes that the subject experiences 20 fewer asthma exacerbations per week. When considering how to treat future patients, the doctor may wonder: Did she make the right choice in prescribing drug 1, or should she have prescribed drug 2?

Perhaps we might say yes, she did make the right choice, since the patient’s asthma exacerbation rate was reduced. But what if we knew that, had the patient received drug 2, his weekly asthma exacerbation count would have decreased by 50 instead of the observed 20? In that case, the best action would have been  $a^{(2)}$ . The difficulty is that the doctor can never truly observe this counterfactual scenario where she selected  $a^{(2)}$ , in order to compare it to what actually happened. We can never truly know what *would have happened* if we had done something else; Causal Inference requires us to reason about two outcomes while conceding that the data can only ever show us one. To the scientist who may despair at this notion I offer a quote from the absurdist philosopher Albert Camus [31]: “when life reaches a certain temperature, our souls and our blood . . . live at ease in contradiction, as indifferent to duty as to faith.” And so like Camus’ intrepid portrayal of Sisyphus, we return once more to our boulder of inference, estimating potential outcomes cheerfully and in full acceptance of how unachievable the utopia of observing them is.

In the above example, the 20 point exacerbation reduction and the 50 point exacerbation reduction are *potential outcomes*, and they are counterfactual to one another in that only one of these outcomes can truly occur. The 20 point reduction is the effect caused by  $a^{(1)}$ , and the 50 point reduction is the effect caused by  $a^{(2)}$ . Determining causal effects by estimating potential outcomes is the fundamental principle of Causal Inference. Figure 2.1 demonstrates this heuristic; here, the dashed line represents the unchosen (counterfactual) path, and the solid line represents the chosen one. The potential outcomes framework was developed by Donald Rubin in the 1970s [136]; we will discuss the broader scientific landscape at the time later in Chapter 6.

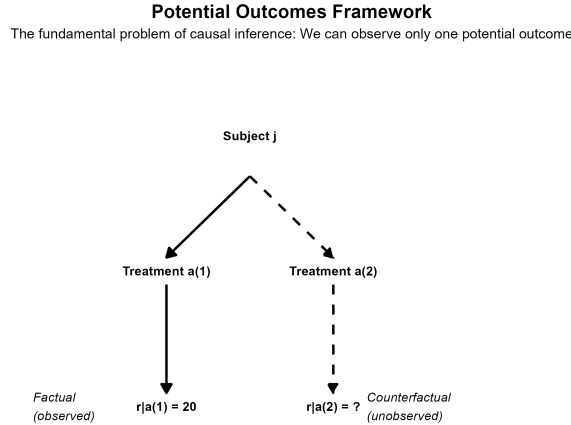


FIGURE 2.1: Illustration of the potential outcomes framework.

Let us use  $R$  to denote an arbitrary asthma rate reduction, with specific values  $R = r$ , and let  $A$  denote an arbitrary action. We will also use  $\pi$  to denote the action selection policy, where  $\pi(A)$  returns the probability of selecting action  $A$  from some action set  $\mathcal{A}$ . In the asthma example  $\mathcal{A}$  would be a set containing two elements,  $\mathcal{A} = \{a^{(1)}, a^{(2)}\}$ , and the probability of selecting treatment 2 would be denoted  $\pi(A = a^{(2)})$ . Using this notation, the conditional outcome  $r|a$  given an action  $a \in \mathcal{A}$  for the example above is given in Equation (2.1).

Typically, Causal Inference notation would use  $Y$  instead of  $r$  for the outcome, and denote the outcome under action 1 as  $Y^{a=1} = 20$  (with  $Y^{a=2} = 50$  under action 2).

Also, in Causal Inference, the action-selection policy  $\pi$  is called the *treatment-assignment mechanism*. I prefer “policy” because it is both more concise and more general, since actions may not be treatments in contexts other than clinical trials.

$$r|A = \begin{cases} 20 & A = a^{(1)} \\ 50 & A = a^{(2)} \end{cases} \quad (2.1)$$

**Definition 2.1 (Causal Risk, Average Treatment Effect)** *The causal risk of applying an action  $a$  is the outcome caused by applying action  $a$ . The causal risk difference between two actions  $a^{(1)}, a^{(2)}$  is  $r|a^{(2)} - r|a^{(1)}$ , the difference in outcomes due to the differing actions. In the above example the causal risk difference is  $50 - 20 = 30$ . Similarly, the causal risk ratio is  $\frac{r|a^{(2)}}{r|a^{(1)}}$ , here  $\frac{50}{20}$ .*

*The Average Treatment Effect (ATE) is defined as the expected causal risk difference between the treatment and control conditions across the entire population (Equation 2.2).*

$$ATE = \mathbb{E}[R|a^{(2)} - R|a^{(1)}] \quad (2.2)$$

Note that we can never compute the ATE directly without some type of estimation, because direct computation would require all subjects to appear in both treatment arms. We cannot ever know what would have happened to a specific subject had he been treated with the drug he did not actually receive. Figure 2.1 illustrates this concept: For each subject  $j$ , we can only observe the outcome under the treatment actually received (solid lines). The counterfactual outcome under the alternative treatment remains unobserved (dashed lines). In this example, the subject receives treatment  $a^{(1)}$  and experiences an outcome  $r|a^{(1)} = 20$ , but we cannot observe what would have happened under treatment  $a^{(2)}$ .

TABLE 2.1: Hypothetical observed data from four patients.

| Subject  | $A$       | $r a^{(1)}$ | $r a^{(2)}$ | CRD    |
|----------|-----------|-------------|-------------|--------|
| Julius   | $a^{(1)}$ | 20          | ?           | ? - 20 |
| Octavian | $a^{(1)}$ | 25          | ?           | ? - 25 |
| Claudius | $a^{(2)}$ | ?           | 50          | 50 - ? |
| Augustus | $a^{(2)}$ | ?           | 45          | 45 - ? |

One can, however, make counterfactual estimates about *populations* of subjects. Suppose, instead of one subject, the doctor observes 4 subjects with asthma. She gives 2 of them drug 1 (i.e. applies action  $A = a^{(1)}$ ) and gives the rest of them drug 2 ( $A = a^{(2)}$ ), observing the data in Table 2.1. One way to compare the causal effects of drug 1 and drug 2 is by estimating the mean and variance of the ATE, and using these values for a significance test. If Table 2.1 had no missing data, we might simply use the average of the Causal Risk Difference (CRD) column as an estimate of the ATE, under the assumption that the CRD column  $\{\text{CRD}_j\} = \{? - 20, ? - 25, 50 - ?, 45 - ?\}$  is an i.i.d. Normal random sample, i.e.  $\{\text{CRD}_j\} \sim \text{MVN}(ATE, \text{diag}(\sigma^2))$ .

Let  $J_k$  be the set of all subject indices such that subject  $j$  was assigned to treatment group  $k$ , e.g.  $J_1 = (\text{Julius}, \text{Octavian})$ , with  $J = |J_1| + |J_2| = 2 + 2 = 4$ . Let  $M$  be the missing data indicator matrix where  $M_{jk} = 1$  if  $j \in J_k$  (indicating that  $R_j|a^{(k)}$  is observed) and 0

TABLE 2.2: Hypothetical data from four patients with counterfactuals imputed.

| Subject  | $A$       | $r a^{(1)}$ | $r a^{(2)}$ |
|----------|-----------|-------------|-------------|
| Julius   | $a^{(1)}$ | 20          | 47.5        |
| Octavian | $a^{(1)}$ | 25          | 47.5        |
| Claudius | $a^{(2)}$ | 22.5        | 50          |
| Augustus | $a^{(2)}$ | 22.5        | 45          |

otherwise. We can fill in (impute) the missing datapoints in Table 2.1 under the assumption that within each column, the missing items do not differ substantially from the non-missing items (Equation 2.3). This condition is also referred to as *exchangeability* (see Definition 2.2). Exchangeability means that the subjects who receive action  $a^{(1)}$  are interchangeable with those who receive  $a^{(2)}$  with respect to their potential outcomes. Under exchangeability, the treatment groups are comparable, and we can interpret observed differences in outcomes as causal effects.

$$\mathbb{E}[R|a^{(2)}] - \mathbb{E}[R|a^{(1)}] = \mathbb{E}[R|a^{(2)}, M_{j2} = 1] - \mathbb{E}[R|a^{(1)}, M_{j1} = 1] \quad (2.3)$$

**Definition 2.2 (Exchangeability)** *A treatment assignment policy  $\pi$  satisfies exchangeability if the potential outcomes are independent of the actual treatment assignment. Formally, for all actions  $a \in \mathcal{A}$ :*

$$A \perp \{R|a : a \in \mathcal{A}\} \quad (2.4)$$

When conditioning on covariates  $X$ , we have conditional exchangeability:

$$A \perp \{R|a : a \in \mathcal{A}\} \mid X \quad (2.5)$$

*Randomization* is one way to ensure that exchangeability holds. To see why exchangeability allows us to estimate potential outcomes from observed data, consider the following derivation. The quantity  $\mathbb{E}[R|a^{(2)}]$  is a population-level potential outcome: the expected response if *everyone* received treatment 2. We wish to show that under exchangeability, this equals  $\mathbb{E}[R|a^{(2)}, M_{j2} = 1]$ , the expected outcome among subjects who were actually assigned to treatment 2, i.e. the quantity we can compute from the data we observe.

$$\mathbb{E}[R|a^{(2)}] = \mathbb{E}[R|a^{(2)}, M_{j2} = 1] P(M_{j2} = 1) + \mathbb{E}[R|a^{(2)}, M_{j2} = 0] P(M_{j2} = 0) \quad (2.6)$$

$$= \mathbb{E}[R|a^{(2)}, M_{j2} = 1] P(M_{j2} = 1) + \mathbb{E}[R|a^{(2)}, M_{j2} = 1] P(M_{j2} = 0) \quad (2.7)$$

$$= \mathbb{E}[R|a^{(2)}, M_{j2} = 1] \underbrace{(P(M_{j2} = 1) + P(M_{j2} = 0))}_{=1} \quad (2.8)$$

$$= \mathbb{E}[R|a^{(2)}, M_{j2} = 1] \quad (2.9)$$

Equation (2.6) is the law of total expectation, which decomposes the expected value by observation status (that is, by whether subject  $j$ 's outcome under treatment 2 was observed ( $M_{j2} = 1$ ) or counterfactual ( $M_{j2} = 0$ )). In Equation (2.7), exchangeability allows us to replace the counterfactual expectation  $\mathbb{E}[R|a^{(2)}, M_{j2} = 0]$  with the observable one  $\mathbb{E}[R|a^{(2)}, M_{j2} = 1]$ , since the potential outcomes do not depend on group assignment. Note that this result holds for *any* randomization ratio, not just 50/50, because the probabilities  $P(M_{j2} = 1)$  and  $P(M_{j2} = 0)$  sum to one regardless of their individual values.

If exchangeability holds, and if we assume that each group-specific outcome vector  $\{r|a^{(k)}\}_{J_k}$  is i.i.d. Normal, then we can impute the missing values and estimate the ATE by computing the mean over the CRD column in Table 2.2. An R implementation of this simple ATE estimation procedure follows.

#### Estimating CRD and ATE

```
# toy_ate_crd.R
# Incomplete potential outcomes data
df <- data.frame(
  Subject = c("Julius", "Octavian", "Claudius", "Augustus"),
  Treatment = c("a1", "a1", "a2", "a2"),
  R_a1 = c(20, 25, NA, NA), # Observed only for a1 group
  R_a2 = c(NA, NA, 50, 45) # Observed only for a2 group
)

# Impute missing values using group-wise means
mean_r_a1 <- mean(df$R_a1, na.rm = TRUE)
# mean for those who received a1
mean_r_a2 <- mean(df$R_a2, na.rm = TRUE)
# mean for those who received a2

df$R_a1[is.na(df$R_a1)] <- mean_r_a1
df$R_a2[is.na(df$R_a2)] <- mean_r_a2

# Compute causal risk difference for each subject
df$CRD <- df$R_a2 - df$R_a1

# Estimate ATE
ate <- mean(df$CRD)

# Output
print(df)
cat("\nEstimated ATE (R_a2 - R_a1):", round(ate, 2), "\n")
```

Running the above code, we see that the estimated ATE is 25.

|   | Subject  | Treatment | R_a1 | R_a2 | CRD  |
|---|----------|-----------|------|------|------|
| 1 | Julius   | a1        | 20.0 | 47.5 | 27.5 |
| 2 | Octavian | a1        | 25.0 | 47.5 | 22.5 |
| 3 | Claudius | a2        | 22.5 | 50.0 | 27.5 |
| 4 | Augustus | a2        | 22.5 | 45.0 | 22.5 |

Estimated ATE (R\_a2 - R\_a1): 25

**To the Reinforcement Learning expert:** Exchangeability is important in Reinforcement Learning as well. Without exchangeability, methods like importance sampling for off-policy evaluation would give biased estimates (see [77]), because the distribution of states visited under the behavior policy would systematically differ in their reward potential from what the target policy would encounter. We will return to these con-

cepts, drawing explicit connections between weighting strategies in Causal Inference and Reinforcement Learning, in future chapters.

In addition to exchangeability, there are two additional assumptions upon which the potential outcomes estimation framework, as described above, depends. First, we require that each subject's outcome depends only on his or her own treatment assignment, and that each treatment has a single well-defined version. This is known as the Stable Unit Treatment Value Assumption (SUTVA).

**Definition 2.3 (Stable Unit Treatment Value Assumption (SUTVA))** *The Stable Unit Treatment Value Assumption is satisfied iff the two following conditions hold:*

1. **No interference:** *The potential outcome of any subject  $j$  under action  $a$  does not depend on the treatment assignments of other subjects. That is,  $R_j|a$  is unaffected by whether some other subject  $j' \neq j$  receives action  $a^{(1)}$  or  $a^{(2)}$ .*
2. **No hidden variations of treatment:** *For each action  $a \in \mathcal{A}$ , there is only one version of that treatment. If two subjects both receive action  $a^{(1)}$ , they receive the same treatment.*

SUTVA was formalized by Rubin [139] and is assumed throughout this book. In practice, SUTVA violations can arise in settings such as vaccine trials (where treating one subject may protect others through herd immunity, violating no-interference) or when treatment delivery varies across clinical sites (violating the single-version requirement). See [69] for further discussion.

Second, we require the *consistency* assumption, which allows one to connect the potential outcomes notation to observable data.

**Definition 2.4 (Consistency)** *If a subject  $j$  receives treatment  $A_j = a$ , then the observed outcome  $R_j$  equals the potential outcome under that treatment:*

$$A_j = a \implies R_j = R_j|a \quad (2.10)$$

Without the consistency assumption, there is no link between the potential outcomes we reason about and the data we actually observe. In particular, the derivations in Equations 2.6-2.7 require consistency to hold. The ATE estimation procedure that follows also requires that the data are consistent. In practice, consistency is an assumption that is generally taken for granted, but rarely mentioned.

Finally, identification of causal effects also requires *positivity*: every subject must have a nonzero probability of receiving each treatment, i.e.  $\pi(A = a) > 0$  for all  $a \in \mathcal{A}$ . Under randomization, positivity holds by design, since the randomization ratio assigns every arm with positive probability. Positivity becomes a substantive concern in observational settings, where some subgroups may never receive certain treatments; we will formalize the notion of positivity in Chapter 3.

When any of the assumptions above (exchangeability, SUTVA, consistency, and positivity) fail, then our models may be inaccurate, even if they appear reasonable. This is because, without the assumptions, probability begins to behave in ways that defy our intuition. A classic illustration of what happens when assumptions are left implicit comes from the Monty Hall problem, an infamous probability puzzle based on a television game show. In the game show, a contestant picks one of three doors, behind one of which is a car; the other two conceal goats. After the contestant's initial pick, the host (who knows what is behind each door) opens one of the remaining doors to show a goat. The contestant may then

switch to the other unopened door or stay with their original choice. The counterintuitive result is that switching gives a 2/3 probability of winning the car, while staying gives only 1/3.

Most clinical problems are more complex than this situation, but with some cartoonish reductions, one can engineer an equivalent situation, as I have written about in a recent article in *Significance* [20]. While the specific reduction is a cartoonish oversimplification, the point is that a similar type of unintuitive behavior may still arise from unmet assumptions in more complex, realistic situations, though it is less easily demonstrable.

The cartoon version goes like this: Suppose a psychiatrist believes that three medications (call them A, B, C) are equally likely to be the best option for a patient with ADHD. She initially prescribes drug A, after first checking the patient's chart for contraindications to it and finding none. Before the patient fills the prescription, a further chart review reveals a pre-existing condition (say, low blood pressure) that rules out drug C due to side effects. Should the psychiatrist switch the patient to drug B? By the same logic as the game show problem, yes: The probability that drug B is optimal is now approximately 2/3, while drug A remains at 1/3.<sup>1</sup>

To see why, and to see where the assumptions matter, we can write the problem in potential outcomes notation. Let  $y_i$  be the (continuous) quality-of-life improvement the patient would experience on drug  $i$ , for  $i \in \{A, B, C\}$ . Suppose the expected improvement is

$$\mu_i = \beta_e x_{e,i} - \beta_s x_{s,i} \quad (2.11)$$

where  $x_{e,i}$  is the efficacy of drug  $i$ ,  $x_{s,i}$  is the severity of its side effects for this patient, and  $\beta_e, \beta_s > 0$  are coefficients. The optimal drug is  $\arg \max_i \mu_i$ . Three conditions are required for the Monty Hall logic to apply:

1. **Exactly one treatment is optimal:**  $\mu_i \neq \mu_j$  for all  $i \neq j$ . Since  $\mu_i$  depends on continuous patient characteristics, ties have probability zero, so this holds generically.
2. **Information arrival does not reveal the optimal treatment.** A pre-existing condition can inflate  $x_{s,i}$  (making drug  $i$  worse), but it cannot tell us which drug is best. In game show terms, the host can only open a door with a goat behind it.
3. **Information arrives about an unselected treatment.** The chart review reveals something about drug C, not about drug A (which was already selected). In the clinical setting, this is guaranteed by the pre-screening step: Because the psychiatrist has already checked drug A for contraindications before prescribing it, any contraindication discovered afterward must concern one of the unselected drugs. This condition corresponds to the game show rule that the host never opens the door the contestant has selected.

One objection I received goes like this: “The doctor could just as easily discover something that rules out drug B, not drug C.” But if the pre-existing condition ruled out drug B, and drug B was truly optimal, then drug B was not actually optimal for this patient once safety is accounted for. Optimality in Equation (2.11) already incorporates both efficacy *and* side effects. An equivalent objection in the game show would be: “What if Monty opened

<sup>1</sup>Strictly speaking, the *exact* value of 2/3 requires one further feature of the game show: the host *always* opens a door. In the clinical setting, whether a contraindication exists to be discovered is up to nature. After we condition on the fact that one *was* discovered, the probability that drug B is optimal lies between 1/2 and 2/3, depending on how likely non-optimal drugs are to carry discoverable contraindications.  $P(B \text{ is optimal})$  equals 2/3 in the limit where such a contraindication is certain to surface. Switching remains advantageous throughout this range, so the qualitative conclusion is unaffected.

the door with the car behind it?” The rules of the game forbid that. Here, the “rules” are the structure of medical information: a discovered contraindication makes a drug *worse* for the patient, never better.

There is also another cartoonish reduction here. The Monty Hall logic requires that a single pre-existing condition cannot rule out two drugs at once. If one condition (say, severe renal impairment) eliminated both B and C simultaneously, only drug A would remain and there would be nothing to switch to. This amounts to a constraint on the action space: the set of available treatments must have at least two members after the information update.

Finally, the initial prescription must be uninformative about which drug is truly best. If the psychiatrist already had reason to believe drug A was superior and selected it on that basis, the uniform prior (1/3 each) would not hold, and the switching advantage would be attenuated or eliminated. This corresponds, in our earlier language, to a form of exchangeability: the initial assignment is random with respect to the potential outcomes.

Note that in this analogy, it is *nature itself* who plays the role of the “game show host”, not the doctor. The doctor is the contestant, making treatment selections under uncertainty. The “host”, nature, opens doors by revealing the patient’s pre-existing conditions, genetic makeup, and medical history, which exist independently of the doctor’s choices and whose revelation follows its own probabilistic logic. The doctor “plays against” nature in the sense of statistical decision theory, where one selects a decision rule (a treatment policy) to perform well against an unknown state of the world. Formulations of this type, where a scientist or clinician chooses a policy to optimize outcomes against a stochastic environment, have been studied in the statistical treatment rules literature [103] and have direct analogues in the Reinforcement Learning framework developed in Part II of this book.

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## 2.2 Hypothesis Testing and the Randomized Controlled Trial

Since the ATE in the asthma example above was positive, we might be tempted to conclude that drug 2 is better. But there is a problem: Since values are truly continuous, it is very unlikely that the ATE will be exactly zero; some difference in ATE is always expected due to random fluctuations in the data, even if the drugs have exactly the same efficacy levels. Is the positive ATE a result of differing drug efficacy levels, or is it simply random chance that treatment 2 came out on top?

We can answer this question by fitting a statistical model to the vector of outcomes  $\{r\}$  as described in Equation (2.12), where  $\varepsilon_j$  is assumed to be drawn from some probability distribution. The simplest assumption would be that  $\varepsilon_j \sim N(0, \sigma^2)$  with  $\sigma^2$  assumed constant across all  $j$ .

$$r_j | a_j^{(k)} = \theta_k + \varepsilon_j, k \in \{1, 2\} \quad (2.12)$$

Throughout this book, when we write  $X \sim D$ , we mean that we are assuming  $X$  behaves as if it had been drawn from the distribution  $D$ . The phrase  $\varepsilon_j \sim N(0, \sigma^2)$ , then, translates to “ $\varepsilon_j$  was drawn from a normal distribution with mean 0 and variance  $\sigma^2$ ”. Readers unfamiliar with distributional notation might visit Chapter 1 of [32] for a review. The following sections will also assume basic knowledge of probability density/cumulative distribution functions, expected values, and variance, which unfamiliar readers may review in Chapters 2-3 of [32].

The choice of  $\varepsilon_j \sim N(0, \sigma^2)$  is an assumption which will underlie most of the causal models discussed herein. Note that this is an *assumption*, not a discovered fact, just as Equation 2.12 is a model that the analyst chooses. Whether or not these assumptions are



“correct” is generally not an answerable question; the statistician can typically only promise that the assumption is *reasonable* within the context provided. Indeed, some distributional choices do have specific justifications; for example, the central limit theorem gives us reasons to expect a Normal distribution when the variable arises from the sum of many small independent contributions. This is plausibly the case in the asthma example, though strictly speaking exacerbation counts are bounded (may not be less than zero) and discrete (we cannot have 2.5 exacerbations), so in practice a model which could account for these things - such as Poisson or negative binomial - would be a more faithful starting point. However, for our illustrative purposes, the Gaussian is simpler and does often provide sufficiently accurate estimations in practice.

If treatment groups are exchangeable, then the ATE between groups  $k = 1$  and  $k = 2$  is  $\text{ATE} = \theta_2 - \theta_1$ . Moreover, exchangeability allows us to estimate the ATE using  $\widehat{\text{ATE}} = \hat{\theta}_2 - \hat{\theta}_1$ . If subjects are assigned to groups randomly *via* the toss of an unbiased coin, then the treatment arms are exchangeable. The key takeaway is that **randomization implies exchangeability**, which consequently allows Causal Inference.

The *Randomized Controlled Trial* (RCT) is the formalized system which the healthcare industry usually uses to perform this process of drug comparison. These RCTs can be understood through the lens of Aristotelian logic. In his *Rhetoric* [5], Aristotle describes how there are three parts to any message: Sender, subject and receiver. The receiver can take on one of three forms:

1. The *observer*, who makes inference about the accuracy of the data source,
2. The *juror*, who draws conclusions about past events, and
3. The *judge*, who makes decisions about future events.

To establish the efficacy of a new drug, we assume each of these three roles in succession: First, we gather data (the message) and act as the observer when we perform quality control to ensure that the data are accurate. Next, we act as the juror; we analyze the data and draw some conclusions about what happened. Based on these conclusions, we then make a decision about how to treat future patients, acting as the judge, as we perform a statistical *hypothesis test*. Critically, the judge’s role assumes that the future will be like the past, i.e. that the patients whose data we collect are analogous to future patients which we will see in clinical practice.

**To the Reinforcement Learning practitioner:** The Aristotelian framework maps directly to the exploration-exploitation cycle. One would enter observer status when in exploration phase, collecting reliable data under randomization, then act as the juror when evaluating said data by estimating Q-values or advantages. One acts as the judge when shifting to exploitation phase, deploying the learned policy by selecting greedy actions. The assumption that “the future will be like the past” corresponds to the stationarity assumption in MDPs.

The statistical hypothesis test is likely a well-known concept to the RL practitioner; indubitably it is second-nature to the statistician. The hypothesis test is itself a type of causal question, though the causality is with regards to an abstract parameter rather than a physical causal effect. The hypothesis test asks the following: *Is the difference between  $\hat{\theta}_1$  and  $\hat{\theta}_2$  i) solely due to the random fluctuations in the data, or ii) due to differing treatment efficacy levels?* In case i),  $\theta_1 = \theta_2$ , so the difference between groups is entirely due to the variance on  $\{\varepsilon_j\}$ . This first case i) is also known as the *null hypothesis* or  $H_0$ . Conversely, in case ii), which is also known as the *alternative hypothesis* or  $H_a$ , the difference would have

two causes: One of these is again the random vector  $\{\varepsilon_j\}$ . The second cause is that  $\theta_1 \neq \theta_2$ . The statistical hypothesis test dichotomizes these two cases into counterfactuals that cannot simultaneously exist; it cannot be true that both  $\theta_1 = \theta_2$  and  $\theta_1 \neq \theta_2$ . But unlike the counterfactuals considered in Section 2.1, which prognostically forecasted different futures under different potential selectable actions, the hypothesis test looks *backward* from the data and considers potential *causes* (signal or noise) rather than future outcomes (Figure 2.2).

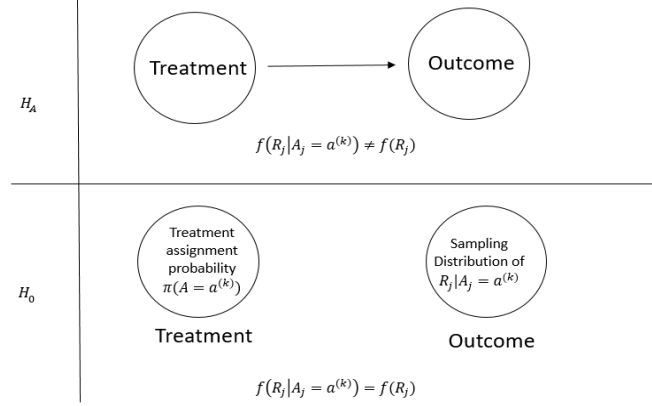


FIGURE 2.2: Causal diagram illustrating the hypothesis testing framework.

In the 1953 psychology classic “Science and Human Behavior” [158], B.F. Skinner discusses the ways we frequently construct both the forward-looking and retroactive causal frameworks in our everyday lives. If we know that someone has not drunk water for several days, for example, then we might assume that he is very thirsty; forward-looking inference, as in the potential outcomes framework, would forecast that should he have access to water he would drink. However, causes are sometimes also inferred from their downstream effects: If we see someone drinking copious amounts of water, we might naturally assume that this person is very thirsty, ascribing the thirst cause retroactively. The person’s thirst, in this case, corresponds to the statement  $\theta_1 \neq \theta_2$ . The hypothesis test, then, is analogous to this second case, where we infer thirst from the drinking behavior. Modern Reinforcement Learning methods have strong parallels to Skinner’s work; this intersection between optimal control theory and behaviorist psychology is discussed in Chapter 14 of [164].

When observing a person drinking copious amounts of water, our inference of thirst is dependent on an assumption that were the person not thirsty, the probability that he would drink so much water is very low. We compare  $H_0$  with  $H_a$  the same way. The classical framework compares expected outcomes under each potential causal scenario we could be in.

Note that this causal interpretation of hypothesis testing separates the parameter entity from the physical force that the parameter represents. If exchangeability does not hold, then the parameter may cause the difference in groups, but this parameter does not represent the physical effect we wish to learn about. Figure 2.3 illustrates this distinction: Under exchangeability, treatment assignment causally determines the parameter value, which in turn causes the outcome. When exchangeability is violated, the parameter still affects out-

comes, but treatment assignment does not determine the parameter; instead, unmeasured confounders drive both treatment selection and outcomes. Note that in both cases the population parameter of the outcome is different between groups ( $p < .05$ ), but only under exchangeability does this statistical relationship reflect the causal effect of treatment.

At this point, I invite you to pause here to ponder what “ $p < \alpha$ ” actually guarantees. The  $\alpha$  level is a long-run frequency: it says that across many hypothetical replications of the same trial under the null hypothesis, we would falsely reject in an  $\alpha$  fraction of them.  $\alpha$  tells us the reliability of the *tool* we are using, not the probability that we have falsely rejected the null in our particular study. *Power* (denoted  $1 - \beta$ , where  $\beta$  is the Type II error rate), which we will introduce in the subsequent paragraph, is the same kind of object: A long-run rejection rate under the alternative. Neither  $\alpha$  nor  $\beta$  tells us anything specific about the particular trial in front of us. Beginners often find this counterintuitive: they look at their freshly randomized clinical trial, notice that the smokers happen to have ended up disproportionately in the treatment arm, and conclude that something has gone wrong. Surely we should adjust the analysis to fix this, they think. And while it is true that covariate adjustment is in many cases a good idea, the reasoning comes from the *variance* of the estimator. Covariate adjustment does not “fix” the random imbalance, because there is nothing to fix; the long-run guarantees of randomization already hold by construction; covariate adjustment simply improves the efficiency of the analysis.

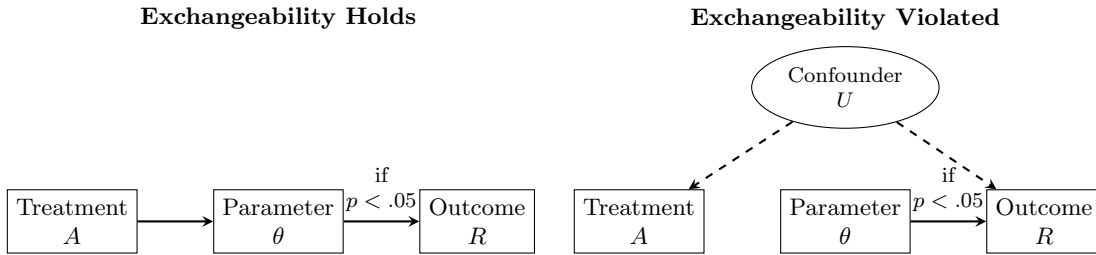


FIGURE 2.3: Causal interpretation of exchangeability in hypothesis testing. Left: Under exchangeability, the parameter represents the true treatment effect. Right: When exchangeability is violated, an unobserved confounder influences both treatment assignment and the outcome; consequently, the estimated parameter  $\theta$  no longer represents the causal effect of treatment.

If exchangeability holds, then the *power* of a statistical test (i.e. the probability of rejecting  $H_0$  when a true difference exists) depends on the sample size. With sufficient sample size, we can obtain a  $p$ -value less than  $\alpha$  with probability approaching 1, since the chances that the two group means will be exactly equal is zero. The question of when to stop sampling is therefore of utmost importance.

There is also an ethical reason why one should randomize the minimum number of subjects possible in a clinical trial: If there are true differences in treatment efficacy, then as long as subjects are assigned treatments randomly, some subjects will be randomized to the worse treatment arm, which is unethical if it can be avoided. It is better to have the study stop as soon as it is possible to detect the difference between groups so that all subsequent patients can be assigned to the better treatment arm.

**To the Reinforcement Learning practitioner:** The minimum sample size consideration in RCTs can be thought of as a type of exploration-exploitation tradeoff. Every randomized, exploratory action that chooses a suboptimal arm represents poten-

tial harm (regret). Algorithms like UCB and Thompson Sampling attempt to minimize this regret while still learning effectively; the same goal as optimal sample size determination in clinical trials. The difference is that adaptive algorithms can update continuously rather than waiting for a single hypothesis test. We will return to this in Chapter 4.

Many statistical methods exist for the purpose of determining the minimum number of samples required for some test. A thorough discussion of sample size estimation in clinical trials is available in [75].

Most real-life problems require more sophisticated statistical models than the two sample test. The linear model is a flexible and popular tool commonly used by statisticians in Causal Inference problems, particularly those deployed in clinical trials.

**To the Reinforcement Learning practitioner:** Linear models are just linear function approximators with closed-form solutions. Indeed, the approximation of  $Q(s, a)$  with  $\theta^T \phi(s, a)$  in linear TD learning may be done by using `lm()` in R. In fact, `lm()` also provides additional information, namely uncertainty quantification through standard errors.

In maximum likelihood estimation (MLE), we find the parameter value that maximizes, retrospectively, the probability of observing our data. Consider, first, a single observation from our sample,  $r_1$ . We assume  $r_1 \sim \mathcal{N}(\mu, \sigma^2)$ , where  $\mu$  is defined according to 2.13. The parameter set that we wish to estimate, therefore, is  $\Theta = \{\theta_1, \theta_2, \sigma^2\}$ .

$$\mu = \begin{cases} \theta_1 & \text{if subject } j \text{ received } a^{(1)} \\ \theta_2 & \text{if subject } j \text{ received } a^{(2)} \end{cases} \quad (2.13)$$

In practice, however, it is more mathematically convenient to use the slightly less-intuitive parameterization in 2.14, assuming that the goal is to determine which treatment is better. Note that  $\beta_1 = \theta_2 - \theta_1$  and  $\beta_0 = \theta_1$ ;  $\beta_1$  is the treatment effect that we can gain actionable insights from. However, it is easy to move between parametric forms using the beta-theta transformation supplied below.

$$\mu = \begin{cases} \beta_0 & \text{if subject } j \text{ received } a^{(1)} \\ \beta_0 + \beta_1 & \text{if subject } j \text{ received } a^{(2)} \end{cases} \quad (2.14)$$

We switch from using  $\theta$  to  $\beta$  here per statistical convention. Typically, statisticians will use  $\Theta$  to denote abstract parameter sets (individual parameters may be denoted  $\theta$ ).  $\beta$  is usually used specifically for linear model coefficients. So, in the abstract a  $\theta$  might be a linear model coefficient, a variance parameter on some random variable, or something else. In statistical convention, it would be strange to see the notation  $\beta$  used in this way.

If the outcomes  $r$  are normally distributed, then the distribution of  $R_1$  before it is observed is given by the function  $f(r) = \frac{1}{\sigma\sqrt{2\pi}} \exp\left(-\frac{(r-\mu)^2}{2\sigma^2}\right)$ . This function  $f(r)$  is a *probability density function* (PDF).

**Definition 2.5 (Probability Density Function (PDF))** *The probability density function (PDF) of a continuous random variable  $R$  is the derivative of its cumulative distribution function (CDF)  $F(r) = P(R \leq r)$  with respect to  $r$  (see Equation 2.15).*

$$f(r) = \frac{d}{dr}F(r) \quad (2.15)$$

Note that the PDF is defined as a derivative. Hence, the PDF exists only for continuous random variables. Also, the PDF does not give the probability of observing  $R = r$  exactly (which is zero for any continuous variable), but rather describes the density of probability near  $r$ . Probabilities are obtained by integrating:  $P(a \leq R \leq b) = \int_a^b f(r) dr$ . For discrete random variables, the analogous object is the probability mass function (PMF),  $p(r) = P(R = r)$ , which *does* assign probabilities directly to each possible exact value.

Since the asthma example technically involves discrete exacerbation counts, a PMF-based model (such as Poisson or negative binomial) would be more faithful; we proceed with the continuous Gaussian formulation for simplicity, consistent with the assumption above. After  $r_1$  is observed, we can plug its actual value into  $f(r)$  to see how “unexpected” of an observation it is; how far out in the tails of  $f$  the actual value of  $r_j$  falls. Similarly, if  $r_2$  is independent of  $r_1$  and also normally distributed, then the probability of observing  $(R_1, R_2)$  before they are recorded would be  $f(r_1) \times f(r_2)$  (since independent probabilities multiply). We repeat this process for all of the observations in the sample, then plug the actual values into each distribution: For example, if we observed the values  $(5, 7)$ , then we could check  $\frac{1}{\sigma\sqrt{2\pi}} \exp\left(-\frac{(5-\mu)^2}{2\sigma^2}\right) \times \frac{1}{\sigma\sqrt{2\pi}} \exp\left(-\frac{(7-\mu)^2}{2\sigma^2}\right)$  to see how likely it would be to observe our data under certain parameters  $\mu, \sigma$ . We say this is the *likelihood* of the observed data. The idea is to find parameters  $(\hat{\mu}, \hat{\sigma})$  to maximize this likelihood.<sup>2</sup> For further explanation, see any undergraduate (or graduate) statistics textbook; the aforementioned text in [32] is a personal favorite. We will provide some examples of ML estimation using the function `lm()` in section 3.1.

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## 2.3 The Ladder of Causation

Judea Pearl and collaborators formalized a hierarchical structure of inquiry to understand where Causal Inference fits within a broader hierarchy of reasoning. This structure is called the *Ladder of Causation* [118], a conceptual framework that organizes different types of questions into three progressively more powerful levels. Each rung on this ladder corresponds to a different kind of query we might pose about the world, each of which requires different kinds of information in order to be answered.

### *Level 1: Association (Seeing)*

At the base of the ladder is **association**, which refers to statistical relationships observed in the data. Questions at this level are of the form *What does seeing observation  $X$  tell me about the outcome  $R$ ?* Methods which can be used to answer this question include correlations, regressions, and conditional probabilities like  $P(R = r \mid X = x)$ . Association is about pattern-matching, and cannot tell us anything about causality.

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<sup>2</sup>For i.i.d. normally distributed outcomes, the maximum likelihood estimates can be algebraically derived; it turns out that these MLEs are also the values which minimize the MSE between observed and predicted values; the MLEs are the least squares estimates. In general this is not always the case.

Many classical Machine Learning models operate at this level: Models such as multilayer perceptrons and Support Vector Machines learn patterns from observed data and make predictions about future observations, assuming the data-generating process remains the same. However, in practice, they are frequently used off-label in an attempt to draw conclusions about other levels of inference. This is an incorrect application of Machine Learning, and it should be avoided if possible. A thorough discussion of classical Machine Learning techniques is available in [191].

One should not fall into the mistaken belief, however, that classical Machine Learning models are valueless because they do not operate at higher levels of the causal ladder. When prediction itself is the goal, association-level reasoning is the appropriate level of analysis, and a great deal of commercial data science operates here for good reason. A spam filter, for example, does not need to know whether the words “free” and “cash” *cause* a message to be spam; the filter needs only to know that they are reliably associated with spam messages, leaving the psychological analysis to the interested behavioral researcher. Similarly, a demand forecasting model for a retailer is not in the business of asking what would happen if the customer’s income were higher; such a model need only report what total sales next quarter are likely to be, given everything observable today. The error is in using a model trained on associations to make causal statements. In the asthma treatment case, for example, we need to know if the treatment *causes* the outcome. For example, suppose we notice that in clinical records from a hospital, sicker patients tend to receive higher doses of medicine. This association does not tell us anything useful about how well the medicine works, or give us any advice about how to optimally treat future patients. The “incorrect application” of Machine Learning mentioned above refers to this kind of category error, not to predictive modeling as a discipline.

### *Level 2: Intervention (Doing)*

On the second rung of Pearl’s ladder, we ask, *What happens to  $R$  if I do  $A$ ?* This rung, *intervention*, is the realm of controlled experiments and Causal Inference. In intervention-level studies, we simulate or observe the effects of actively *changing* a variable, rather than passively observing it, thereby learning something about causality. In the clinical example, differences in outcome between treatment groups would be intervention-level effects, while a difference in outcome between smokers and non-smokers would be an association-level effect, because we do not assign subjects to smoking status groups.

To differentiate an intervention-level variable from an associative one, Causal Inference notation typically uses the “do-operator”, where the probability of some outcome  $Y$  given that we took action  $X$  would be written as  $P(Y \mid \text{do}(X))$ ; in this book’s notation we would write  $P(R \mid \text{do}(a))$ , distinguishing passively observed variables  $X$  from controlled variables  $a$  as discussed. Note that in order to compute  $R \mid \text{do}(a)$ , we need to know how  $a$  was assigned; we need to know exactly what *do* means.

The policy notation developed in this chapter is the Reinforcement Learning version of the do operator. The policy function  $\pi$  is a generalized roadmap for selecting actions: Recall that  $\pi(A = a \mid X)$  would return the probability of selecting action  $a$  given that we observed  $X$ . If  $\pi$  is deterministic (selecting actions with probability 1), then sometimes researchers will define it to return the selected action instead of a probability, so that  $\pi(X) = a$ . The do-operator  $\text{do}(A = a')$  is equivalent to a degenerate deterministic policy  $\pi(A = a) = \mathbb{1}(a = a')$ , where  $\mathbb{1}$  is the indicator function. In this book, we will use policy notation since it is more general.

In medicine, Randomized Controlled Trials (RCTs) operate at the intervention level by actively assigning treatments and measuring outcomes. However, as we will see in later chapters, many Causal Inference methods also allow us to estimate intervention effects from

observational data using assumptions about the underlying data-generating process.

### Level 3: Counterfactuals (Imagining)

Recall, for a moment, the following line of inquiry which we posed in earlier sections:

Octavian received drug 1 and improved slightly. But what would have happened if we had given him drug 2 instead?

This hypothetical reasoning about what *would have happened* is an example of *counterfactual reasoning*, which sits at the top of Pearl’s hierarchy. Methods which allow us to reason about counterfactual potential universes can assist with solving complex problems requiring more advanced reasoning and analogical thinking.

In Causal Inference, counterfactuals are typically written in the form  $P(Y_{x'} = y' \mid X = x, Y = y)$ , which expresses the probability that  $Y$  would have been  $y'$  if we had done  $x'$ , even though in reality we observed  $X = x$  and  $Y = y$ ; note that Causal Inference notation uses  $x$  for both the action and the covariate. In the notation of this book, we translate this to  $P((R_j|a') = r' \mid A_j = a, R_j = r, X_j = x)$ , where  $a' \neq a$ . Conditioning on the observed treatment and outcome ( $A_j = a, R_j = r$ ) is what distinguishes this Level 3 counterfactual from the Level 2 interventional quantity  $P(R \mid \pi(A = a') = 1)$ .

To answer counterfactual questions, we often need structural models that describe the mechanisms by which they interact; mere associations do not suffice. The conventional RCT is able to estimate counterfactuals on a population level, but subject-level treatment effects require more sophisticated modelling methods, which we will discuss in future chapters.

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## 2.4 Further Reading

Further theoretical and technical development is available in Bareinboim’s foundational textbook mentioned previously [11]. Rubin’s seminal paper *Estimating causal effects of treatments in randomized and nonrandomized studies* [136], which introduces the potential outcomes framework and establishes randomization as a sufficient condition for exchangeability, underpins much of modern Causal Inference theory. *Statistics and Causal Inference* by Holland [65] offers a philosophical and formal treatment of causality using the potential outcomes framework, with an especially clear discussion of what it means for causes to be “active.” More recently, a comprehensive and rigorous treatment of the potential outcomes approach, ideal for statisticians interested in both randomized trials and observational studies is available in [69]. *Sample Sizes for Clinical Trials* [75] by Julious is a modern practical reference which provides tools and formulae for calculating the number of subjects needed in randomized trials, and offers valuable insight into the ethical and statistical tradeoffs discussed in the final section of this chapter.